```
> restart;#Flux Electromag force aprox;
```

```
> de := 1/c^2·H(r)·diff(u(t), t, t) + u(t)·diff(H(r), r, r) = 0;
```

$$de := \frac{H(r)\left(\frac{d^2}{dt^2}\,u(t)\right)}{c^2} + u(t)\left(\frac{d^2}{dr^2}\,H(r)\right) = 0 \tag{1}$$

```
> dsolve(de, u(t));
```

$$u(t) = _C1 \sin\left(\frac{c\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) + _C2 \cos\left(\frac{c\sqrt{\frac{d^2}{dr^2}\,H(r)}\;t}{\sqrt{H(r)}}\right) \tag{2}$$

```
>  eq1 := r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r) = 0;
```

$$eq1 := r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r) = 0 \tag{3}$$

```
> eq2 := (r·diff(H3(r), r, r) + diff(H3(r), r) + (ν^2·r - b^2·r^2/R)·H3(r))/r = 0;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,H3(r)\right) + \frac{d}{dr}\,H3(r) + \left(\nu^2 r - \frac{b^2 r^2}{R}\right) H3(r)}{r} = 0 \tag{4}$$

```
> # Thus with R:=3.566·10^22 m  we can  ignore the therm -b^2·r/R for
  1/R < r < R;
```

$$\frac{1}{R} < r < R \tag{5}$$

```
> 1/r diff((r·diff H(r), r), r) + k^2· H(r) = 0;
```

$$\frac{2\,diff\left(\frac{d}{dr}\,H(r)\right) + r\,diff\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2 H(r) = 0 \tag{6}$$

```
> de1 := 1/r·diff(r·diff(H(r), r), r) + k^2·H(r);
```

$$de1 := \frac{\frac{d}{dr}\,H(r) + r\left(\frac{d^2}{dr^2}\,H(r)\right)}{r} + k^2 H(r) \tag{7}$$

```
> bc := H(2 R) = 0, D(H)(1/R) = R ;
```

$$bc := H(2\,R) = 0,\ D(H)\left(\frac{1}{R}\right) = R \tag{8}$$

```
> dsol := dsolve({de1, bc}, H(r));
```

(9)

$$dsol := H(r) = \frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \quad (9)$$

```
> H(r) := (R BesselY(0, 2 k R) BesselJ(0, k r)) / (k (BesselJ(0, 2 k R) BesselY(1, k/R) - BesselY(0, 2 k R) BesselJ(1, k/R)))
  - (BesselJ(0, 2 k R) R BesselY(0, k r)) / (k (BesselJ(0, 2 k R) BesselY(1, k/R) - BesselY(0, 2 k R) BesselJ(1, k/R)));
```

$$H := r \mapsto \frac{R\cdot\mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} - \frac{\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot R\cdot\mathrm{BesselY}(0, k\cdot r)}{k\cdot\left(\mathrm{BesselJ}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\cdot k\cdot R)\cdot\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \quad (10)$$

```
> u(t) := C1 sin(c sqrt(d^2/dr^2 H(r)) t / sqrt(H(r))) + C2 cos(c sqrt(d^2/dr^2 H(r)) t / sqrt(H(r)));
```

$$u := t \mapsto C1\cdot\sin\left(\frac{c\cdot\sqrt{\frac{\mathrm{d}^2}{\mathrm{d}r^2}H(r)}\cdot t}{\sqrt{H(r)}}\right) + C2\cdot\cos\left(\frac{c\cdot\sqrt{\frac{\mathrm{d}^2}{\mathrm{d}r^2}H(r)}\cdot t}{\sqrt{H(r)}}\right) \quad (11)$$

```
> Hm := subs(H(r) = (R·BesselY(0, 2·k·R)·BesselJ(0, k·r)) / (k·(BesselJ(0, 2·k·R)
  ·BesselY(1, k/R) - BesselY(0, 2·k·R)·BesselJ(1, k/R)))
  - (BesselJ(0, 2·k·R)·R·BesselY(0, k·r)) / (k·(BesselJ(0, 2·k·R)
  ·BesselY(1, k/R) - BesselY(0, 2·k·R)·BesselJ(1, k/R))), H(r)·u(t)/r);
  # Because we have Effective Wave Function
```

$$Hm := \frac{1}{r}\left(\left(\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right.\right. \tag{12}$$

$$\left.-\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\,\mathrm{BesselY}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right)$$

$$\left(C1\,\sin\left(\left(c\left(\right.\right.\right.\right.$$

$$\left.\left.\left.\left.-\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\left(\mathrm{BesselJ}(0,k\,r)-\frac{\mathrm{BesselJ}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\right.\right.\right.\right.$$

$$\left.\left.\left.\left.+\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\left(\mathrm{BesselY}(0,k\,r)-\frac{\mathrm{BesselY}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\right)^{1/2}t\right)\right)\Big/$$

$$\left(\frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \right.$$

$$\left. - \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \right)^{1/2}$$

$$+ C2\cos\left(\left(c\left(\right.\right.\right.$$

$$\left. - \frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\left(\mathrm{BesselJ}(0, k\,r) - \frac{\mathrm{BesselJ}(1, k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right.$$

$$\left. \left. \left. + \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\left(\mathrm{BesselY}(0, k\,r) - \frac{\mathrm{BesselY}(1, k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)} \right)^{1/2} t \right) \Bigg/$$

$$\left(\frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \right.$$

$$\left. \left. - \frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)} \right)^{1/2} \right) \right) \right)$$

```
> Fmag0 := subs( t = 0, Hm/(2·Pi·r·l) );
```

$$Fmag0 := \frac{1}{2\,r^2\,\pi\,l}\left(\left(\frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)}\right.\right. \tag{13}$$

$$-\frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)}\Bigg)$$

$$(C1\sin(0) + C2\cos(0))\Bigg)$$

```
> solve(Fmag0 = 0);
```

Warning, solutions may have been lost

$$\{C2 = 0, R = R, k = k, l = l, r = r\},\ \{C2 = C2, R = RootOf(-\mathrm{BesselJ}(0, 2\,k\,_Z)\,\mathrm{BesselY}(0, k\,r) + \mathrm{BesselJ}(0, k\,r)\,\mathrm{BesselY}(0, 2\,k\,_Z)), k = k, l = l, r = r\} \quad (14)$$

```
> Fmag := subs(t = 1, Hm/(2·Pi·r·l));
```

$$Fmag := \quad (15)$$

$$\frac{1}{2\,r^2\,\pi\,l}\Bigg(\Bigg(\frac{R\,\mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)}$$

$$-\frac{\mathrm{BesselJ}(0, 2\,k\,R)\,R\,\mathrm{BesselY}(0, k\,r)}{k\left(\mathrm{BesselJ}(0, 2\,k\,R)\,\mathrm{BesselY}\left(1, \frac{k}{R}\right) - \mathrm{BesselY}(0, 2\,k\,R)\,\mathrm{BesselJ}\left(1, \frac{k}{R}\right)\right)}\Bigg)$$

$$\Bigg(C1\sin\Bigg(\Bigg(\Bigg(c\Bigg($$

$$-\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\left(\mathrm{BesselJ}(0,k\,r)-\frac{\mathrm{BesselJ}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}$$

$$+\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\left(\mathrm{BesselY}(0,k\,r)-\frac{\mathrm{BesselY}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\Bigg)^{1/2}\Bigg)\Bigg/$$

$$\left(\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right.$$

$$\left.-\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\,\mathrm{BesselY}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right)^{1/2}\Bigg)$$

$$+C2\cos\Bigg(\Bigg(\Bigg(c\left(\right.\right.\right.$$

$$-\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\left(\mathrm{BesselJ}(0,k\,r)-\frac{\mathrm{BesselJ}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}$$

$$+\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\left(\mathrm{BesselY}(0,k\,r)-\frac{\mathrm{BesselY}(1,k\,r)}{k\,r}\right)k}{\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)}\Bigg)^{1/2}\Bigg)\Bigg/$$

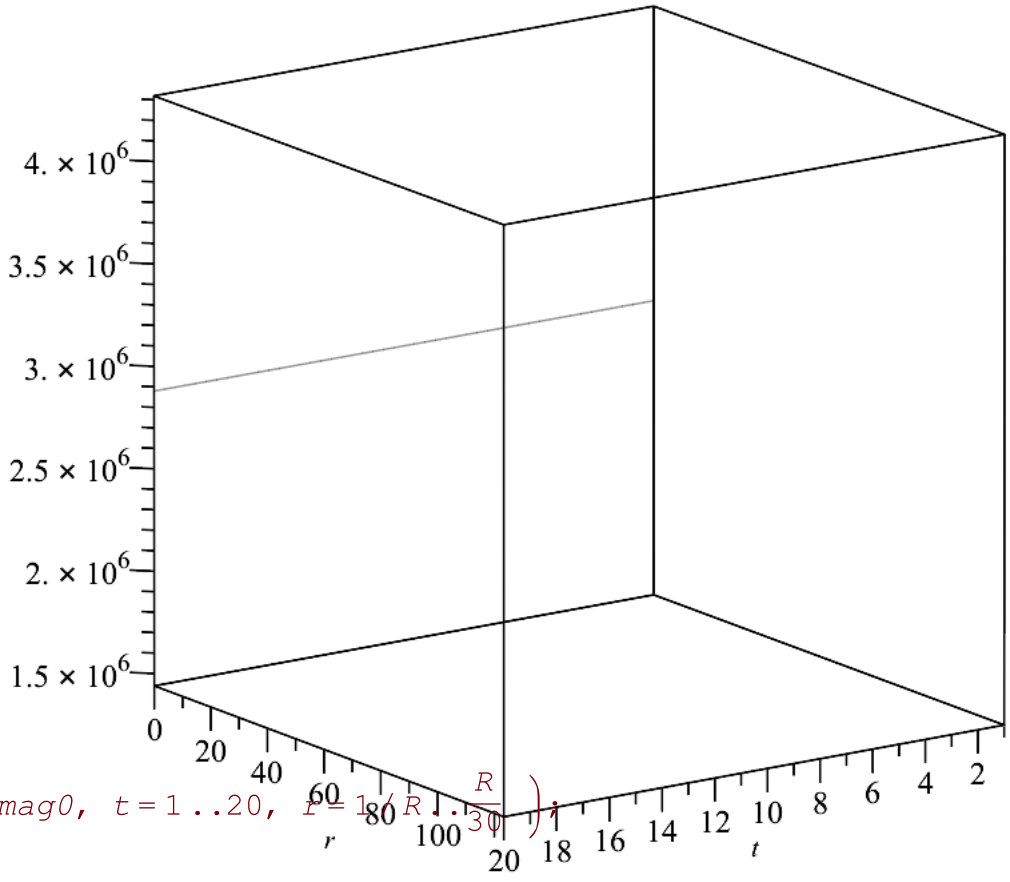
$$\left(\frac{R\,\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right.$$

$$\left.-\frac{\mathrm{BesselJ}(0,2\,k\,R)\,R\,\mathrm{BesselY}(0,k\,r)}{k\left(\mathrm{BesselJ}(0,2\,k\,R)\,\mathrm{BesselY}\left(1,\frac{k}{R}\right)-\mathrm{BesselY}(0,2\,k\,R)\,\mathrm{BesselJ}\left(1,\frac{k}{R}\right)\right)}\right)^{1/2}\Bigg)\Bigg)\Bigg)$$

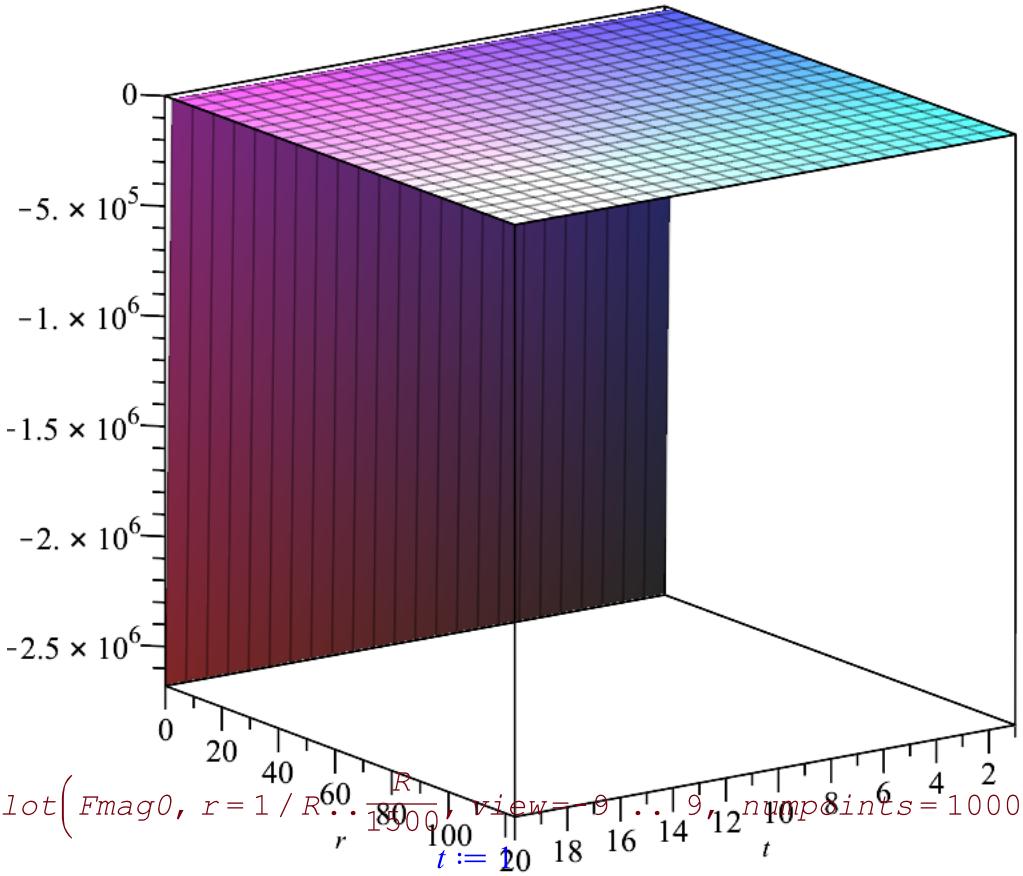
```
> R := 3.7·10^3;  C1 := 1; C2 := 1; k := 1; c := 30; l := 8;
```

$$R := 3700.0$$
$$C1 := 1$$
$$C2 := 1$$
$$k := 1$$
$$c := 30$$
$$l := 8 \qquad \textbf{(16)}$$

```
> plot3d( Fmag, t = 1..20, r = 1/R..R/30 );
```

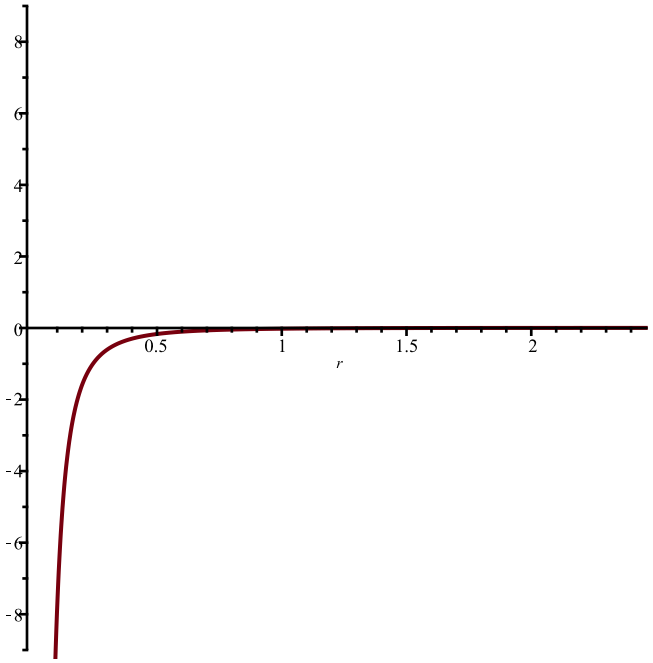



```
> plot3d( Fmag0, t=1..20, r=1/R..R/30 );
```



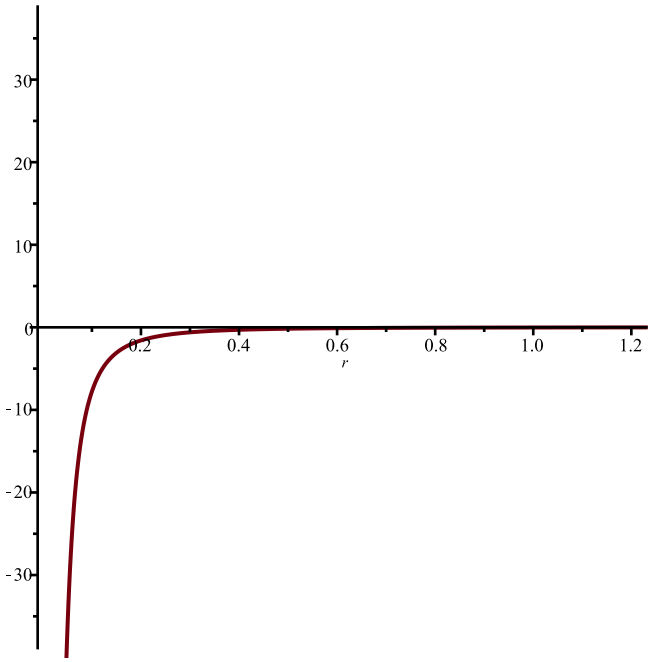

```
> t := 1; plot( Fmag0, r = 1 / R .. R/1000, view=-0 .. 9, numpoints=1000 );
```

t := 1



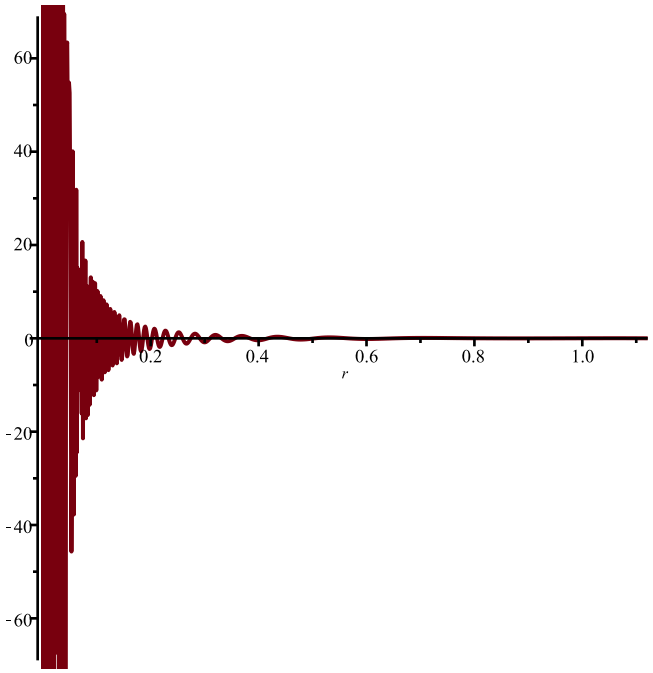

```
> t := 0; plot( Fmag0, r = 1 / R .. R/3000, view=-39 .. 39, numpoints=1000 );
```

$t := 0$




```
> t := 0; plot(Fmag, r = 1/R .. R/3300, view=-69 .. 69, numpoints=1000);
```

$t := 0$




```
> #For dimensional calculus =>bc:=H(2·R) = 0, D(H)(a/R) = b·R;
    # where dim [b] =[L^(-3/2)] = 1  meter^(-3/2); dim[a]=[L^2] ; a = 1; dim[H]
    =[L^(-1/2)] because dim[H(r)]=dim[r·f(r)]=[L^(-1/2)];
> #with unitar value a=1, b=1
```

